

STUDENT NUMBER

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2011

TRIAL HIGHER SCHOOL CERTIFICATE

EXAMINATION

Mathematics Extension 1

General Instructions

- Reading time – 5 minutes
- Working time – 2 hours
- Write using black or blue pen
- Board-approved calculators may be used
- A table of standard integrals is provided at the back of this paper
- All necessary working should be shown in every question

Total marks – 84

- Attempt Questions 1–7
- All questions are of equal value

QUESTION	MARK
1	/12
2	/12
3	/12
4	/12
5	/12
6	/12
7	/12
TOTAL	/84

THIS QUESTION PAPER MUST NOT BE REMOVED FROM THE EXAMINATION ROOM

This assessment task constitutes 40% of the Higher School Certificate Course Assessment.

Total marks – 84

Attempt Questions 1–7

All questions are of equal value

Answer each question in a SEPARATE writing booklet. Extra writing booklets are available.

Question 1 (12 marks) Use a SEPARATE writing booklet.

Marks

- (a) Evaluate $\lim_{x \rightarrow 0} \frac{\sin 4x}{3x}$ **1**
- (b) Differentiate $\tan^{-1}(\sin x)$ **2**
- (c) Use the table of standard integrals to find $\int \frac{1}{\sqrt{x^2 - 16}} dx$ **1**
- (d) Solve the inequation : $\frac{3x}{2-x} \leq 2$ **3**
- (e) The point P divides the interval AB externally in the ratio $5:2$. **2**
Given that A and B have coordinates $(-4, 2)$ and $(2, -1)$ respectively,
find the coordinates of P .
- (f) Use the substitution $u = 25 - x^2$ to evaluate $\int_3^4 7x\sqrt{25 - x^2} dx$ **3**

Question 2 (12 marks) Use a SEPARATE writing booklet.

Marks

(a) Solve $3^x = 5$.

2

Express your answer correct to two decimal places.

(b) Evaluate: $2 \int_0^{\frac{\pi}{8}} \cos^2 4x dx$

3

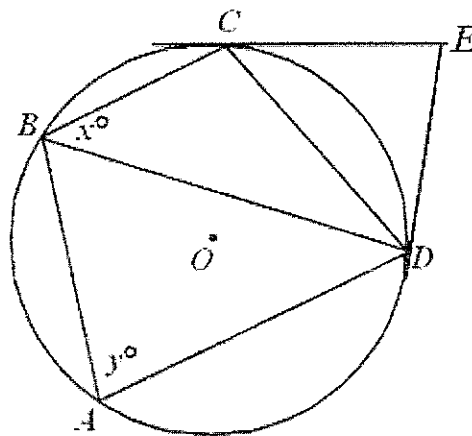
(c) Find the general solution to $\sin 2x + \sin x = 0$.

3

Express your answer in terms of π .

- (d) The circle ABCD has centre O. Tangents are drawn from the external point E to contact the circle at C and D.

$$\angle CBD = x^\circ \text{ and } \angle BAD = y^\circ.$$



(i) Explain why $\angle CED = (180 - 2x)^\circ$.

2

(ii) Show that $\angle BDC = (y - x)^\circ$.

2

Question 3 (12 marks) Use a SEPARATE writing booklet.

Marks

- (a) (i) State the domain and range of the function $f(x) = 2\cos^{-1}\left(\frac{x}{3}\right)$ **2**
- (ii) Sketch the function $f(x) = 2\cos^{-1}\left(\frac{x}{3}\right)$ **1**
- (b) Consider the function $f(x) = x^3 - \log_e(x+1)$
- (i) Show that the function has a root between 0.8 and 0.9. **1**
- (ii) Using the value 0.8 and one application of Newton's method, find a better approximation for this root, correct to two decimal place. **3**
- (c) Gary and George are cooking a 2 kilogram roast, which has an initial temperature of 10°C . They place it in an oven which has been preheated to 190°C at 5:00pm. It is found that the temperature of the roast increases to 50°C after 75 minutes.
- A roast is considered medium rare when it reaches a temperature of 65°C . At time t its temperature T increases according to the equation
- $$\frac{dT}{dt} = -k(T - 190), \text{ where } k \text{ is a positive constant.}$$
- (i) Show that $T = 190 + Ae^{-kt}$ is a solution to the equation, where A is a constant **1**
- (ii) Find the values of A and k **2**
- (iii) At what time will the roast be medium rare? (answer correct to the nearest minute) **2**

Question 4 (12 marks) Use a SEPARATE writing booklet.

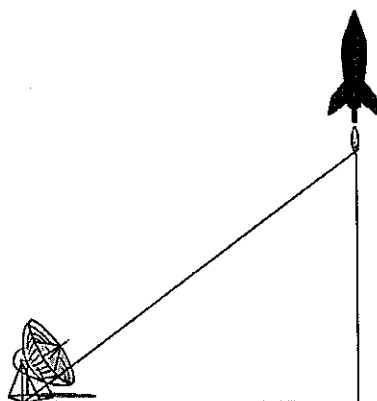
Marks

- (a) Use the principle of mathematical induction to show that $5^n + 2 \times 11^n$ is divisible by 3 for all positive integers n . 3
- (b) The velocity $v \text{ ms}^{-1}$ of a particle moving in a straight line along the x -axis is given by $v^2 = 12 + 4x - x^2$.
- (i) Show that the particle is moving in simple harmonic motion. 2
- (ii) Between which two points is the particle oscillating? 2
- (iii) What is the amplitude of the motion? 1
- (c) When the polynomial $P(x)$ is divided by $(x+2)(x-3)$, the quotient is $Q(x)$ and the remainder is $R(x)$.
- (i) Explain why the most general form of $R(x)$ is given as: $R(x) = ax + b$, where a and b are constants. 1
- (ii) Given that $P(3) = 1$, show that $R(3) = 1$. 1
- (iii) Find $R(x)$ if $P(3) = 1$, $R(3) = 1$ and when $P(x)$ is divided by $(x+2)$ the remainder is 6. 2

Question 5 (12 marks) Use a SEPARATE writing booklet.

Marks

- (a) The function $f(x) = \operatorname{cosec} x$ for $\frac{\pi}{2} \leq x < \pi$
- (i) State the domain of the inverse function $f^{-1}(x)$ 1
- (ii) Show that $f^{-1}(x) = \sin^{-1}\left(\frac{1}{x}\right)$ 2
- (iii) Hence find $\frac{d}{dx} f^{-1}(x)$ 1
- (b) A rocket is launched vertically and is tracked by a radar station, which is located on the ground 3km from the launch site. 3



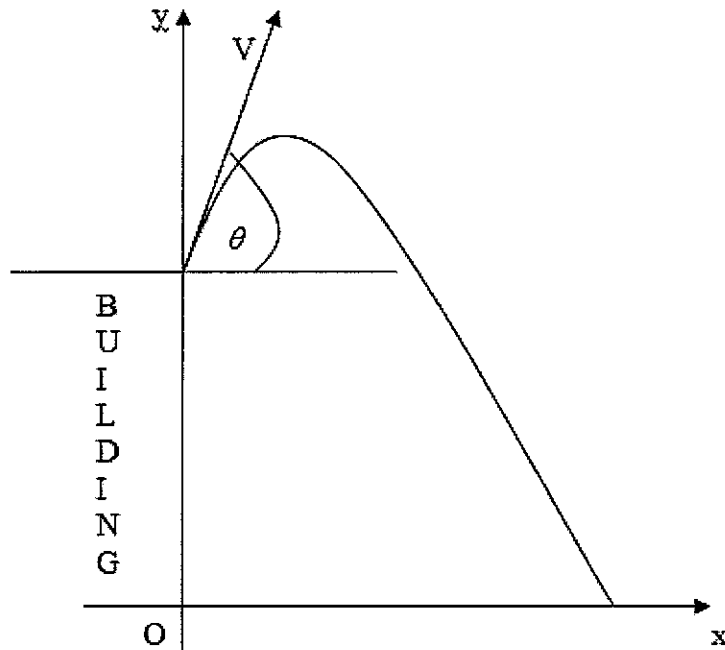
What is the vertical speed of rocket at the instant when its distance from the radar station is 5km and this distance is increasing at the rate of 5000 km per hour?

- (c) (i) Show that the equation of the normal at $R(10q, 5q^2)$ on the parabola $x^2 = 20y$ is given by $x + qy = 5q^3 + 10q$. 2
- (ii) The normal intersects the y-axis at T. Find the coordinates of T and hence show that the coordinates of S, the midpoint of RT, is given by $S(5q, 5q^2 + 5)$. 2
- (iii) Hence find the Cartesian equation of the locus of S. 1

Question 6 (12 marks) Use a SEPARATE writing booklet.

Marks

- (a) Jackson throws a ball from the top of an 18 metre high building with a velocity (V) of 12 m s^{-1} . The angle of projection (θ) is 60° to the horizontal.



Assume that the equations of motion for the ball are $\ddot{x} = 0$ and $\ddot{y} = -10$, and that air resistance is negligible.

- | | | |
|-------|---|---|
| (i) | Let (x, y) be the position of the ball at time t seconds after it was thrown and before it hits the ground. | 3 |
| | Show that $x = 6t$ and $y = -5t^2 + 6\sqrt{3}t + 18$. | |
| (ii) | Calculate the time that has elapsed before the ball hits the ground. | 2 |
| (iii) | What is the maximum height reached by the ball. | 2 |

Question 6 continues on page 8

Question 6 (continued)

(b) If $f(x) = g(x) - \ln[g(x) + 1]$

(i) Prove $f'(x) = \frac{g(x)g'(x)}{g(x) + 1}$. **2**

(ii) Hence evaluate $\int_{\frac{\pi}{12}}^{\frac{\pi}{4}} \frac{\sin 2x \cos 2x}{\sin 2x + 1} dx$. **3**

End of Question 6

Question 7 (12 marks) Use a SEPARATE writing booklet.

Marks

(a) (i) Show that $\tan(rx) \tan(r+1)x = \frac{\tan(r+1)x}{\tan x} - \frac{\tan(rx)}{\tan x} - 1$ 2

(ii) Hence, or otherwise, show that 2

$$\tan 20^\circ \tan 40^\circ + \tan 40^\circ \tan 60^\circ + \dots + \tan 180^\circ \tan 200^\circ = -9$$

- (b) A river running due east has straight parallel banks. A vertical post stands with its base, P , on the north side of the river. On the south bank are two surveyors, A who is to the east of B and B who is to the west of the post.

A & B are at a distance $\frac{2a}{7}$ apart and the $\angle APB = 150^\circ$.

The angles of elevation from A and B to the top Q , of the post are 45° and 30° respectively.

(i) Draw a diagram to show this information.

(ii) Find in terms of a ,

(α) the height of the post. 2

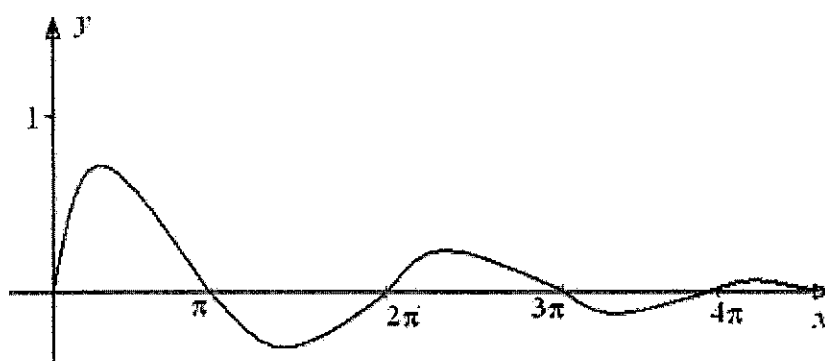
(β) the width of the river 2

Question 7 continues on page 10

Question 7 (continued)

- (b) (i) Show that $y = e^{-x} \sin x$ has infinitely many local maxima/minima. 1

A sketch of the graph of $y = e^{-x} \sin x$, for $x \geq 0$ is given below:



- (ii) The area between the curve $y = e^{-x} \sin x$, the x axis, and the ordinates between $x = a$ and $x = b$ is given by

$$A = \frac{1}{2} \left| \left[e^{-x} (\cos x + \sin x) \right]_a^b \right| \quad (\text{You do not need to prove this})$$

- (α) Using the formula given above, prove that the area enclosed between the curve, the x axis, and the ordinates $x = 0$ and 1

$$x = \pi \text{ is } \frac{1}{2} (e^{-\pi} + 1).$$

- (β) Find similar expressions (as shown in (ii)(α)) for the successive areas enclosed between the curve $y = e^{-x} \sin x$ and the x axis and show that 2

$$\text{the sum of those areas for } x \geq 0 \text{ is } \frac{1}{2} \left(\frac{e^{\pi} + 1}{e^{\pi} - 1} \right).$$

End of paper

Extension 1

$$a) \lim_{x \rightarrow 0} \frac{\sin 4x}{3x} = \frac{4}{3} \lim_{x \rightarrow 0} \frac{\sin 4x}{4x} = \frac{4}{3}$$

$$b) \tan^{-1}(\sin x) = \frac{1}{1 + \sin^2 x} \cdot \cos x$$

at this point $\sin$ is differentiated. $\rightarrow$ no need to go on.

$$= \frac{\cos x}{1 + \sin^2 x}$$

$$c) \int \frac{1}{\sqrt{x^2 - 4^2}} dx = \ln(x + \sqrt{x^2 - 16}) + C$$

$$d) \frac{3x}{2-x} \leq 2, \quad x \neq 2$$

$$(2-x)^2 \frac{3x}{2-x} \leq 2(2-x)^2$$

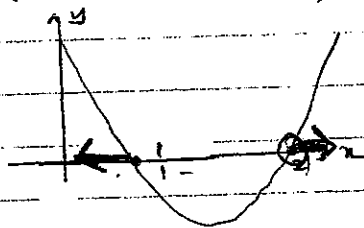
$$3x(2-x) \leq 2(4 - 4x + x^2)$$

$$6x - 3x^2 \leq 8 - 8x + 2x^2$$

$$0 \leq 5x^2 - 14x + 8$$

$$\therefore 0 \leq (5x-4)(x-2)$$

$$\therefore x \leq \frac{4}{5}, x > 2$$



OR $\frac{3x}{2-x} \leq 2 \quad \therefore x \neq 2$

If $x < 2$, $3x \leq 4 - 2x$ (If $x > 2$, $3x \geq 2(2-x)$)

$$5x \leq 4$$

$$x \leq \frac{4}{5} \checkmark$$

$$3x \geq 4 - 2x$$

$$5x \geq 4$$

$$x \geq \frac{4}{5}$$

$$\therefore x > 2$$

$$\therefore x \leq \frac{4}{5}, x > 2$$

1 mark.

MUST SHOW ALL WORKING.

1 mark for correct differentiation of inverse trig

1 mark for correct answer

1 mark. many students missed this!

many students missed this!

1 mark for $x \neq 2$ and either x b.s. by $\square$ then or considering $x < 2$ & $x > 2$.

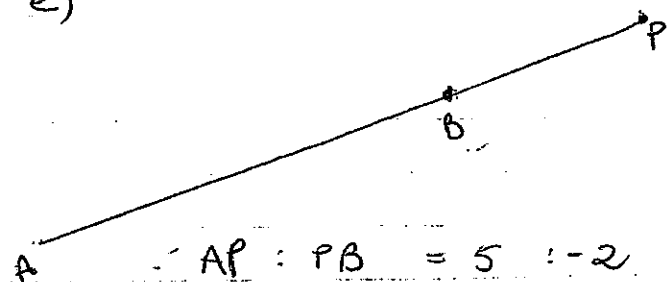
1 mark working. (depending on method.)

1 mark for correct sol'n.

subtract 1 for $x \leq \frac{4}{5}, x \geq 2$

*Students know the method for this question but have little understanding of what they are actually doing

e)



$$\begin{array}{cc} A & B \\ (-4, 2) & (2, -1) \\ 5 & -2 \end{array}$$

$$P\left(\frac{5 \times 2 - 2 \times -4}{5 - 2}, \frac{5 \times -1 + 2 \times 2}{5 - 2}\right)$$

$$= \frac{18}{3}, \frac{-9}{3}$$

$$= (6, -3)$$

This needs reusing for many students

NOT WELL DONE

A wrong formula
= zero marks

1 mark for correct substitution

1 mark for correct answer.

1 mark

Students who found the terminates in terms of 'u' were more successful than those who substituted back for u & used $x = 4 \neq \frac{4}{3}$

1 mark.

This needs reusing for MANY students

1 mark

f) $u = 25 - x^2$ $x = 4, u = 9$
 $du = -2x dx$ $x = 3, u = 16$

$$\int_B^4 7x \sqrt{25 - x^2} dx$$

$$= -\frac{7}{2} \int_9^{16} -2x \sqrt{25 - x^2} du$$

$$= -\frac{7}{2} \int_{16}^9 \sqrt{u} du$$

$$= -\frac{7}{2} \left[\frac{2}{3} u^{3/2} \right]_{16}^9$$

$$= -\frac{7}{3} \left[u^{3/2} \right]_9^{16}$$

$$= -\frac{7}{3} [16^{3/2} - 9^{3/2}]$$

$$= -\frac{7}{3} [4^3 - 3^3]$$

$$= -\frac{259}{3}$$

Question 2

(a) $3^x = 5$

$$x = \log_3 5$$

$$= \frac{\log 5}{\log 3}$$

$$= 1.464973521$$

$$= 1.46$$

$$1 \quad \frac{\log 5}{\log 3}$$

$$1 \quad 1.46$$

(b) $2 \int_0^{\pi/8} \cos^2 4x \, dx = \int_0^{\pi/8} [\cos 8x + 1] \, dx$

1

$$= \left[\frac{1}{8} \sin 8x + x \right]_0^{\pi/8}$$

1

$$= \frac{\pi}{8}$$

(c) $\sin 2x + \sin x = 0$

$$2 \sin x \cos x + \sin x = 0$$

$$\sin x (2 \cos x + 1) = 0$$

$$\sin x = 0, \cos x = -\frac{1}{2}$$

$$x = 0, \pi, \dots, x = \frac{2\pi}{3}, \frac{4\pi}{3}, \dots$$

$$x = n\pi, x = 2n\pi \pm \frac{2\pi}{3}$$

* not knowing expansion
* not factorising

1 for

$$\sin x (2 \cos x + 1) = 0$$

* not know general solution

$$1 \quad \frac{n\pi}{2n\pi \pm \frac{2\pi}{3}}$$

(d) (i) $\angle EDU = x$ (angle in alternate segment is equal to angle between the tangent and chord)

* abbreviated statements
* solution did not

$$EC = ED \text{ (tangents from an external point are equal)}$$

progress (adequate)
* isosceles
just appeared

$$\angle ECD = x \text{ (angles opposite equal sides of an isosceles triangle)}$$

$$\angle CED = 180 - 2x \text{ (angle sum of triangle)}$$

* no such thing as base angle

(ii) $\angle BCD + \angle DAB = 180$ (opposite angles of a cyclic quadrilateral)

$$y + \angle DAB = 180$$

$$\angle DAB = 180 - y$$

$$\angle BDC + x + 180 - y = 180 \text{ (angle sum of triangle)}$$

1

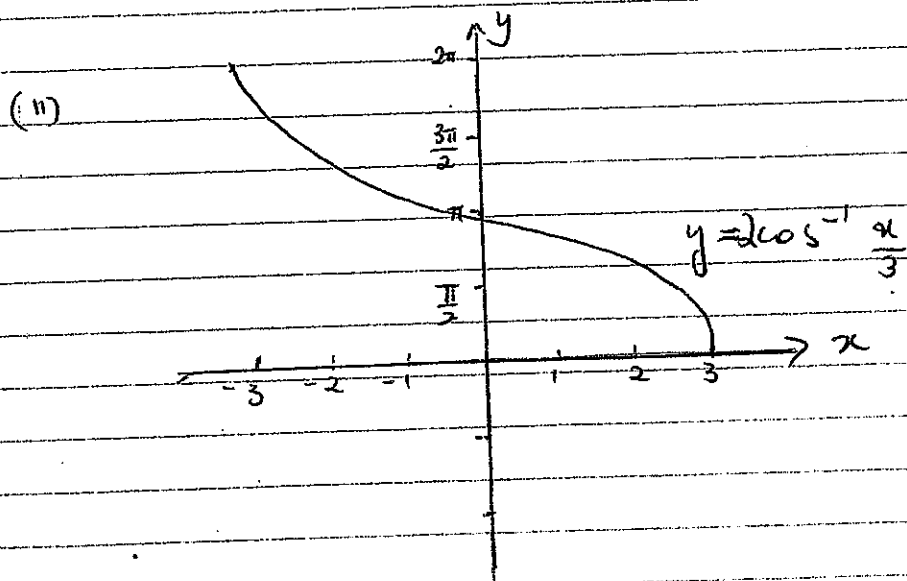
Question 3

$$f(x) = 2 \cos^{-1} \frac{x}{3}$$

(i) $y = \cos^{-1} x$ Domain $-1 \leq x \leq 1$
 $-1 \leq \frac{x}{3} \leq 1$
 $-3 \leq x \leq 3$

$f(x) = \cos^{-1} x$ Range $0 \leq \cos^{-1} x \leq \pi$
 $0 \leq 2 \cos^{-1} x \leq 2\pi$

$\therefore$ Domain $-3 \leq x \leq 3$
 Range $0 \leq y \leq 2\pi$



Most students did this well.

All students should know basic inv. trig curves

1 mark

1 mark.

1 mark.

[must use ruler & have consistent scale]

many students didn't explain the opposite sign!

1 mark, must mention opp sign

students who didn't show subst. often got zero
 1 mark $f(0.8)$

b) (i) $f(x) = x^3 - \ln(x+1)$
 $f(0.8) = 0.8^3 - \ln(1.8) \approx -0.0758...$
 $f(0.9) = 0.9^3 - \ln(1.9) \approx 0.0871...$
 As $f(0.8)$ & $f(0.9)$ are opp sign
 there is a root between 0.8 & 0.9.

(ii) $f(x) = x^3 - \ln(x+1)$ $f(0.8) = -0.0758...$

$f'(x) = 3x^2 - \frac{1}{x+1}$ $f'(0.8) = 3(0.8)^2 - \frac{1}{0.8+1}$
 ≈ 1.364

$x_2 = x_1 - \frac{f(x_1)}{f'(x_1)}$

$= 0.8 - \frac{-0.0758}{1.364}$

$= 0.8555...$

$= 0.86$ (2 dp)

1 mark correct sub into Newton's method

1 mark correct answer rounded (not 0.855... 0.9)

- c) Original temp = 10°C .
surrounding = 190°C .
 50°C after 75 min.

$$(i) T = 190 + Ae^{-kt} \rightarrow Ae^{-kt} = T - 190$$

$$\frac{dT}{dt} = -kAe^{-kt}$$

$$\frac{dT}{dt} = -k(T - 190)$$

$$(ii) T = 190 + Ae^{-kt}$$

When $t=0$, $T=10$

$$\therefore 10 = 190 + Ae^{-k \cdot 0}$$

$$-180 = A$$

$$\therefore A = -180$$

$$\therefore T = 190 - 180e^{-kt}$$

when $t=75$, $T=50$

$$\therefore 50 = 190 - 180e^{-75k}$$

$$-140 = -180e^{-75k}$$

$$\frac{7}{9} = e^{-75k}$$

$$\frac{7}{9} \therefore k = \frac{-\frac{1}{75} \ln \frac{7}{9}}{1} \quad k = 0.003$$

$$(iii) T = 190 - 180e^{-kt}$$

$$65 = 190 - 180e^{-kt}$$

$$-125 = -180e^{-kt}$$

$$\frac{25}{36} = e^{-kt}$$

$$\ln \frac{25}{36} = -kt$$

$$t = \ln \frac{25}{36} \div \frac{1}{75} \ln \frac{7}{9}$$

$$t = 108.8207 \dots$$

$$t = 109 \text{ min (to nearest min)}$$

Roast is ready at 6.49 pm.

Students who integrated has difficulties with $e^x = A$

This was only worth 1 mark

Differentiate $T = 190 + Ae^{-kt}$
1 mark

Well done

1 mark for A

Few transcription errors
READ CAREFULLY!

1 mark for k.

1 mark

Many students neglected to show time
1 mark.

Make sure you
ANSWER THE
QUESTION!

Question 4

(a) For $n=1$

$5^1 + 2 \times 11^1 = 27$ which is divisible by 3.

$\therefore$ true for $n=1$

Assume true for $n=k$.

That is, $5^k + 2 \times 11^k = 3m$ for integer m }
 i.e., $5^k = 3m - 2 \times 11^k$

For $n=k+1$

$$\begin{aligned} & 5^{k+1} + 2 \times 11^{k+1} \\ &= 5^k \cdot 5 + 2 \times 11^{k+1} \\ &= (3m - 2 \times 11^k) 5 + 2 \times 11^{k+1} \\ &= 15m - 10 \times 11^k + 2 \times 11 \times 11^k \\ &= 15m + 12 \times 11^k \\ &= 3 \times (5m + 4 \times 11^k) \text{ which is} \end{aligned}$$

* substituting for 5^k and 11^k
 * trying to fudge result.

Therefore, if it is true for $n=k$ then it is true for $n=k+1$

* no final statement

It is true for $n=1$, $\therefore$ true for $n=1+1=2$

True for $n=2$, $\therefore$ true for $n=2+1=3$
 and so on.

$\therefore$ true for all integer (positive) n .

(b) i $v^2 = 12 + 4x - x^2$

$$\frac{1}{2}v^2 = 6 + 2x - \frac{1}{2}x^2$$

$$\frac{d(\frac{1}{2}v^2)}{dx} = 2 - x$$

$\ddot{x} = -(x-2)$ which is of the

form $a = -\omega^2(x)$ or $\therefore$ acceleration is directly proportional to displacement

* no final statement

Question 4

(b) ii $v^2 = 12 + 4x - x^2$

$$= (6-x)(2+x)$$

$$v = 0 \quad (\text{at the end points})$$

$$\therefore (6-x)(2+x) = 0$$

← 1

$$x = -2 \text{ and } 6$$

$\therefore$ the particle oscillates between $x = -2$ and 6 .

← 1

iii

$$\text{Amplitude} = \frac{8}{2} = 4$$

1

(c) i $P(x) = (x+2)(x-3)Q(x) + R(x)$ * this equation needs to be formed, particularly for ii

$$\frac{P(x)}{(x+2)(x-3)} = Q(x) + \frac{R(x)}{(x+2)(x-3)}$$

The degree of $R(x)$ is less than the degree of the divisor

← 1

$\therefore R(x)$ is of form $ax+b$

(ii) $P(3) = (3+2)(3-3)Q(3) + R(3)$

← 1

$$1 = 0 \times Q(3) + R(3)$$

$$\therefore R(3) = 1$$

iii

$$R(x) = ax+b \text{ and } R(3) = 1$$

$$\therefore 3a+b=1 \quad \text{--- (1)}$$

$$R(-2) = 6$$

$$\therefore -2a+b=6 \quad \text{--- (2)}$$

solving (1) and (2) gives

$$a = -1, \quad b = 4$$

$$\therefore R(x) = -x + 4$$

← 1

* often equations were not formed and thus solved

5. a) (i) Range $f(x) : f(x) \geq 1$

Domain $f^{-1}(x) = x \geq 1$

(ii) $y = \operatorname{cosec} x$

$x = \operatorname{cosec} y$

SHOW

$x = \frac{1}{\sin y}$

$\therefore \sin y = \frac{1}{x}$

$y = \sin^{-1}\left(\frac{1}{x}\right)$

$f^{-1}(x) = \sin^{-1}\frac{1}{x}$

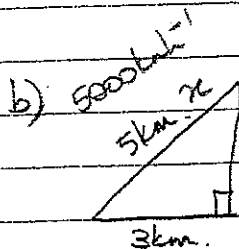
(iii) $\frac{d}{dx} f^{-1}(x) = \frac{d}{dx} \left(\sin^{-1}\left(\frac{1}{x}\right) \right)$

$= \frac{1}{\sqrt{1 - \left(\frac{1}{x}\right)^2}} \cdot -x^{-2}$

$= -\frac{1}{\sqrt{x^2 - 1}} \cdot \frac{1}{x^2}$

$= -\frac{x}{\sqrt{x^2 - 1}} \cdot \frac{1}{x^2}$

$= -\frac{1}{x\sqrt{x^2 - 1}}$



$x^2 = 9 + y^2$

$y = \sqrt{x^2 - 9}$

$\frac{dy}{dx} = \frac{1}{2} (x^2 - 9)^{-\frac{1}{2}} \cdot 2x$

$\frac{dy}{dx} = \frac{x}{\sqrt{x^2 - 9}}$

$\frac{dy}{dt} = \frac{dy}{dx} \cdot \frac{dx}{dt}$

$= \frac{x}{\sqrt{x^2 - 9}} \cdot 5000$

When $x = 5$, $\frac{dy}{dt} = \frac{5}{\sqrt{11}} \cdot 5000 = 6250 \text{ km}^{-1} \text{ 1 mark}$

This was poorly done
Students who took
the time to sketch
1 mark. $y = \operatorname{cosec} x$
did this &
with ease.

1 mark This was a
mark many
students got

This was well done
by most. Students
are reminded that
1 mark SHOW
means all working
must be there.

Students were not
required to simplify
to final answer.

Some students were
saying the $\frac{d}{dx} \left(\frac{1}{x} \right) = \ln x$
Don't confuse
differentiating with
integrating.

1 mark

This was poorly done
Students who drew the
diagram were more
successful.

Students do not seem
to understand that
the only fixed length
1 mark. 15.3

1 mark

$$c) ii) \quad y = \frac{x^2}{20}$$

$$\frac{dy}{dx} = \frac{x}{10}$$

$$\text{at } R(10q, 5q^2) \quad \frac{dy}{dx} = \frac{10q}{10} = q$$

$$m_{\text{tang}} = q$$

$$m_{\text{norm}} = -\frac{1}{q}$$

$$y - 5q^2 = -\frac{1}{q}(x - 10q)$$

$$\text{how } qy - 5q^3 = -x + 10q$$

$$10q + 5q^3 = x + qy$$

$$iii) \quad A+T, \quad x=0$$

$$10q + 5q^3 = 0 + qy$$

$$y = 10 + 5q^2$$

$$T = (0, 5q^2 + 10)$$

$$R(10q, 5q^2)$$

$$S = \left(\frac{10q+0}{2}, \frac{5q^2+5q^2+10}{2} \right)$$

$$\text{how } S = (5q, 5q^2 + 5)$$

$$ii) \quad x = 5q \quad \text{--- (1)}$$

$$y = 5q^2 + 5 \quad \text{--- (2)}$$

$$\text{om (1) } q = \frac{x}{5} \quad y = 5\left(\frac{x}{5}\right)^2 + 5$$

$$y = \frac{x^2}{5} + 5$$

This was well done but again students are reminded that in a SHOW Q, all working must be shown.

1 mark

1 mark

Well done.

1 mark

Well done.

1 mark

Well done.

1 mark

Students were not asked to DESCRIBE the locus, just find the eq'n. To find the focus & vertex & describe

WASTES TIME

Question 6

(a) i Initially



Given $V=12$
 $\theta=60$
 * no setting up of initial conditions.
 * inadequate reasoning/development of results
 $\therefore y\text{-dot} = 12 \times \sin 60 = 6\sqrt{3}$ ← 1

$$\begin{aligned} \ddot{x} &= 0 \\ \dot{x} &= k_1 \\ &= V \cos \theta \\ &= 12 \times \cos 60 \\ \dot{x} &= 6 \\ x &= 6t + k_2 \\ x &= 6t \quad (t=0, x=0) \\ &\quad k_2=0 \end{aligned}$$

$$\begin{aligned} \ddot{y} &= -10 \\ \dot{y} &= -10t + C_1 \\ 6\sqrt{3} &= 0 + C_1, (t=0) \\ \therefore C_1 &= 6\sqrt{3} \\ \dot{y} &= -10t + 6\sqrt{3} \quad \left(\begin{array}{l} \text{1 for } \\ \dot{x} = 6 \end{array} \right) \\ y &= -5t^2 + 6\sqrt{3}t + C_2 \\ 18 &= 0 + 0 + C_2 \\ y &= -5t^2 + 6\sqrt{3}t + 18 \end{aligned}$$

ii

$$\begin{aligned} y &= 0 \\ 5t^2 - 6\sqrt{3}t - 18 &= 0 \\ t &= \frac{6\sqrt{3} \pm \sqrt{468}}{10} \\ &= \frac{3\sqrt{3} \pm 3\sqrt{13}}{5} \\ &\approx 3.2 \text{ seconds} \end{aligned}$$

← 1
 * not understanding relationship between the diagram and the equation y

iii

Max height occurs when $y' = 0$

ie $-10t + 6\sqrt{3} = 0$
 $t = \frac{3\sqrt{3}}{5}$

$$\begin{aligned} \text{Max height} &= -5\left(\frac{3\sqrt{3}}{5}\right)^2 + 6\sqrt{3}\left(\frac{3\sqrt{3}}{5}\right) + 18 \\ &= 23.4 \text{ m} \end{aligned}$$

1 for t

1 (answer)

Question 6

(b) i $f(x) = g(x) - \ln[g(x) + 1]$

$$f'(x) = g'(x) - \frac{g'(x)}{g(x) + 1}$$

$$= \frac{g(x)g'(x) + g'(x) - g'(x)}{g(x) + 1}$$

$$= \frac{g(x)g'(x)}{g(x) + 1}$$

* Those 2 steps need to be given.
← 1

← 1

ii $\int_{\frac{\pi}{12}}^{\frac{\pi}{4}} \frac{\sin 2x \cos 2x}{\sin 2x + 1} dx$

$$= \frac{1}{2} \int_{\frac{\pi}{12}}^{\frac{\pi}{4}} \frac{(\sin 2x)(2)(\cos 2x)}{\sin 2x + 1} dx$$

* Many did not understand the need for $\frac{1}{2}$ and 2
($g(x) = \sin 2x$)
($g' = 2 \cos 2x$)
← 1

$$= \frac{1}{2} \left[\sin 2x - \ln[\sin 2x + 1] \right]_{\frac{\pi}{12}}^{\frac{\pi}{4}}$$

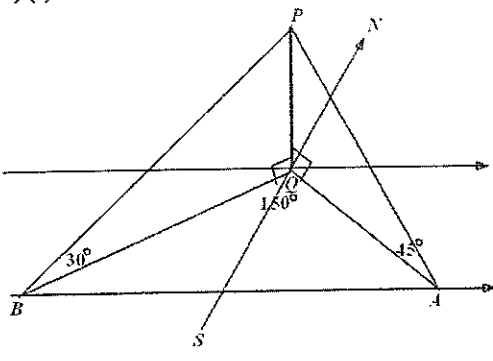
← 1

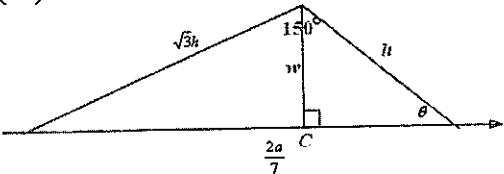
$$= \frac{1}{2} \left\{ \sin \frac{\pi}{2} - \ln(\sin \frac{\pi}{2} + 1) - \left[\sin \frac{\pi}{6} - \ln(\sin \frac{\pi}{6} + 1) \right] \right\}$$

$$= \frac{1}{2} \left\{ 1 - \ln 2 - \frac{1}{2} + \ln \frac{3}{2} \right\}$$

$$= \frac{1}{2} \left(\frac{1}{2} + \ln \frac{3}{2} \right)$$

1 (answer).

a)(i) $\tan x = \tan\{(r+1)x - rx\}$ $= \frac{\tan(r+1)x - \tan(rx)}{1 + \tan(r+1)x \cdot \tan(rx)}$ $\therefore 1 + \tan(r+1)x \cdot \tan(rx) = \frac{\tan(r+1)x - \tan(rx)}{\tan x}$ ie. $\tan(rx) \tan(r+1)x = \frac{\tan(r+1)x}{\tan x} - \frac{\tan(rx)}{\tan x} - 1$	2 marks for correct proof 1 mark: progress towards the solution involving the use of tan (A – B) expansion.	Many students wasted too much time in this question, attempting it 2 or 3 times. As an exam strategy, it is always better to move to the next question which is just an application of this result.
(ii) $\tan 20^\circ \tan 40^\circ + \tan 40^\circ \tan 60^\circ + \dots + \tan 180^\circ \tan 200^\circ =$ $\frac{\tan 40}{\tan 20} - \frac{\tan 20}{\tan 20} - 1 + \frac{\tan 60}{\tan 20} - \frac{\tan 40}{\tan 20} - 1 + \dots +$ $\frac{\tan 200}{\tan 20} - \frac{\tan 180}{\tan 20} - 1 \quad \text{1 mark}$ $= \frac{\tan 200}{\tan 20} - \frac{\tan 20}{\tan 20} - 9 \times 1$ since, $\tan(180 + \theta) = \tan \theta$, $= \frac{\tan 20}{\tan 20} - \frac{\tan 20}{\tan 20} - 9 \times 1$ $= -9$	1 mark: applies the result from (i) 1 mark: simplifies and uses the fact $\tan(180 + \theta) = \tan \theta$ to get the required result.	Many students failed to realise this was an application of the previous question. See in the marking scheme, simply substituting gets you 1 mark.
b)(i) 		Poor drawing skills. <u>Steps to draw a 3D diagram</u> • Draw the river • Draw the North-South line in approximately $22\frac{1}{2}^\circ$. • Erect the post vertically. • Now, complete all the triangles. • Make sure you show right angle sign in the elevation triangles

(ii) Let h be the height of the post In $\triangle PAQ$, $PA = PQ = h$ since $\tan 45 = 1$ In $\triangle PBQ$, $\tan 30 = \frac{PQ}{PB}$ $\therefore PB = \sqrt{3}h$ 1 mark In $\triangle AQB$, using cosine rule, $(\sqrt{3}h)^2 + h^2 - 2\sqrt{3}h^2 \cos 150 = \frac{4a^2}{49}$ $\therefore 3h^2 + h^2 - 2\sqrt{3}h^2 \times \frac{-\sqrt{3}}{2} = \frac{4a^2}{49}$ $7h^2 = \frac{4a^2}{49}$ $\therefore h = \frac{2a}{7\sqrt{7}}$ 1 mark	1 mark: finds the relationship between PA and PB 1 mark: uses cosine rule to evaluate the height of the post.	<i>Some mistakes came in the evaluation process of this cosine rule.</i>
(iii)  In $\triangle PAB$, $\frac{\sin 150^\circ}{\frac{2a}{7}} = \frac{\sin \theta}{\sqrt{3}h}$ $\frac{\sin 150^\circ}{\frac{2a}{7}} = \frac{\sin \theta}{\sqrt{3} \times \frac{2a}{7\sqrt{7}}}$ $\frac{1}{2} = \frac{\sqrt{7} \sin \theta}{\sqrt{3}}$ $\therefore \sin \theta = \frac{\sqrt{3}}{2\sqrt{7}}$ 1 mark In $\triangle PAC$, $\sin \theta = \frac{w}{h}$ $\therefore w = h \sin \theta$ $= \frac{2a}{7\sqrt{7}} \times \frac{\sqrt{3}}{2\sqrt{7}} = \frac{\sqrt{3}a}{49}$ 1 mark	1 mark: Uses sine rule to find an expression for $\sin \theta$ 1 mark: correctly evaluates width of the river.	<i>Another easier approach to this question:</i> Area of the triangle $A =$ $\frac{1}{2} \times (\sqrt{3}h) \times h \sin 150 = \frac{1}{2}$ $\times \frac{2a}{7} \times w$ $W = \left(\frac{(\sqrt{3}h)^2}{4} \right) \times \frac{7}{a}$ Substitute h from (i) gives you $W = \frac{\sqrt{3}a}{49}$

b)(i) $y = e^{-x} \sin x$ $\frac{dy}{dx} = e^{-x} \cos x + \sin x (-e^{-x})$ $= e^{-x} (\cos x - \sin x)$ At stationary points, $\frac{dy}{dx} = 0$. ie. $e^{-x} (\cos x - \sin x) = 0$ since, $e^{-x} \neq 0$, then $\cos x = \sin x$. ie. $\tan x = 0$ $\therefore x = \frac{\pi}{4}, \frac{5\pi}{4}, \frac{9\pi}{4}, \dots$ max.. or min. when $x = \frac{(4n+1)\pi}{4}$, $n = 0, 1, 2, \dots$ ie. infinite number of local maxima.	1 mark: For the proof.	$\cos x - \sin x = 0$ $\cos x = \sin x$ $\tan x = 0$ (not 1 !) was a very common response. You must get the stationary points to get this mark.
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(iii) (α) Area $A_1 =$ $\int_0^{\pi} e^{-x} \sin x dx = \frac{1}{2} [e^{-x} (\cos x + \sin x)]_0^{\pi}$ $= \frac{1}{2} [(1+0) - (-e^{-\pi} + 0)]$ $= \frac{1}{2} (e^{-\pi} + 1)$	1 mark: for proof	Well done
(β) Area $A_2 =$ $\left \frac{1}{2} [e^{-x} (\cos x + \sin x)]_{2\pi}^{\pi} \right $ $= \left \frac{1}{2} [e^{-\pi} (-1+0) - e^{-2\pi} (1+0)] \right $ $= \frac{1}{2} (-e^{-\pi} - e^{-2\pi})$ $= \frac{1}{2} (e^{-2\pi} + e^{-\pi})$ Successive areas will be $A_1 + A_2 + A_3 + A_4 + \dots$ ie. $\frac{1}{2} (e^{-\pi} + 1) + \frac{1}{2} (e^{-2\pi} + e^{-\pi}) +$ $\frac{1}{2} (e^{-3\pi} + e^{-2\pi}) + \frac{1}{2} (e^{-4\pi} + e^{-3\pi}) +$ 	1 mark: Develops the sequence for the sum of areas.	The sequence must be developed to get 1 mark. <u>No pattern can be identified if you are just given the first two terms.</u> This is what a lot of students had done. Gave the first two terms and claimed that it is a GP and surprise, surprise, they can even find the common ratio!

$$\begin{aligned}
&= \frac{1}{2} [(e^{-\pi} + 1) + e^{-\pi}(e^{-\pi} + 1) + \\
&\quad e^{-2\pi}(e^{-\pi} + 1) + e^{-3\pi}(e^{-\pi} + 1) + \dots] \\
&= \frac{1}{2} (e^{-\pi} + 1) [1 + e^{-\pi} + e^{-2\pi} + \\
&\quad e^{-3\pi} + \dots]
\end{aligned}$$

Now, $1 + e^{-\pi} + e^{-2\pi} + e^{-3\pi} + \dots$

is a G.P. where $a = 1$, $r = e^{-\pi}$

Limiting sum exists, since $|r| < 1$.

$$\text{Now, } S_{\infty} = \frac{1}{2} (e^{-\pi} + 1) \left(\frac{a}{1-r} \right)$$

$$= \frac{1}{2} (e^{-\pi} + 1) \left(\frac{1}{1-e^{-\pi}} \right)$$

$$= \frac{1}{2} \left(\frac{e^{-\pi} + 1}{1-e^{-\pi}} \right)$$

$$= \frac{1}{2} \left(\frac{\frac{1}{e^{\pi}} + 1}{1 - \frac{1}{e^{\pi}}} \right)$$

$$= \frac{1}{2} \left(\frac{\frac{1+e^{\pi}}{e^{\pi}}}{\frac{e^{\pi}-1}{e^{\pi}}} \right)$$

$$= \frac{1}{2} \left(\frac{e^{\pi} + 1}{e^{\pi} - 1} \right) \text{ as required}$$

1 mark: applies the sum of GP to get the required result

Fudging was the order of the day with this part of the question!!